

CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

Functional Dependency is an important concept in database design that defines the relationship between attributes in a table. It helps reduce data redundancy and maintain data consistency. In the **SAVANA Online E-Commerce Database Management System**, functional dependencies are analyzed for entities such as Customer, Category, Product, Cart, Orders, Payment, Delivery, and Admin. This analysis helps identify redundancy, prevent data anomalies, and ensure an efficient and well-structured database.

4.2.1 CUSTOMER

Functional Dependency

- Customer_ID → First_Name, Last_Name, Email, Phone_Number, Address, Password

Redundancy Analysis

- Customer details are stored only once using Customer_ID, reducing duplicate customer information.

Data Anomalies

- Insertion Anomaly: Customer details cannot be stored without creating a customer record.
- Update Anomaly: Updating customer email or phone number in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting a customer record removes all customer information.

4.2.2 CATEGORY

Functional Dependency

- $\text{Category_ID} \rightarrow \text{Category_Name, Description}$

Redundancy Analysis

- Category information is stored only once and linked with products.

Data Anomalies

- Insertion Anomaly: A category cannot be created without a Category_ID.
- Update Anomaly: Updating the category name in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting a category may affect related products.

4.2.3 PRODUCT

Functional Dependency

- $\text{Product_ID} \rightarrow \text{Product_Name, Category_ID, Price, Stock, Brand, Color, Size}$

Redundancy Analysis

- Product details are stored only once using Product_ID, avoiding duplicate product information.

Data Anomalies

- Insertion Anomaly: Product details cannot be added without a Product_ID.
- Update Anomaly: Updating product price or stock in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting a product removes all product information.

4.2.4 CART

Functional Dependency

- $\text{Cart_ID} \rightarrow \text{Customer_ID}, \text{Product_ID}, \text{Quantity}, \text{Total_Amount}$

Redundancy Analysis

- Cart details are maintained separately to avoid duplicate cart entries.

Data Anomalies

- Insertion Anomaly: A cart cannot be created without customer and product details.
- Update Anomaly: Updating cart quantity in multiple places may cause inconsistency.
- Deletion Anomaly: Deleting a cart removes all selected products.

4.2.5 ORDERS

Functional Dependency

- $\text{Order_ID} \rightarrow \text{Customer_ID}, \text{Order_Date}, \text{Total_Amount}, \text{Order_Status}$

Redundancy Analysis

- Order information is stored only once and linked with customers.

Data Anomalies

- Insertion Anomaly: An order cannot be created without customer information.
- Update Anomaly: Updating order status in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting an order removes the order history.

4.2.6 PAYMENT

Functional Dependency

- $\text{Payment_ID} \rightarrow \text{Order_ID}, \text{Payment_Method}, \text{Payment_Status}, \text{Payment_Date}$

Redundancy Analysis

- Payment details are stored separately to avoid duplicate payment information.

Data Anomalies

- Insertion Anomaly: Payment cannot be recorded without an order.
- Update Anomaly: Updating payment status in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting a payment record removes transaction history.

4.2.7 DELIVERY

Functional Dependency

- $\text{Delivery_ID} \rightarrow \text{Order_ID}, \text{Delivery_Address}, \text{Delivery_Status}, \text{Delivery_Date}$

Redundancy Analysis

- Delivery details are stored separately, reducing duplicate delivery information.

Data Anomalies

- Insertion Anomaly: Delivery details cannot exist without an order.
- Update Anomaly: Updating delivery status in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting a delivery record removes delivery history.

4.2.8 ADMIN

Functional Dependency

- Admin_ID → Admin_Name, Email, Password

Redundancy Analysis

- Administrator details are stored only once using Admin_ID.

Data Anomalies

- Insertion Anomaly: Admin details cannot exist without an Admin_ID.
- Update Anomaly: Updating admin email or password in multiple records may cause inconsistency.
- Deletion Anomaly: Deleting an admin record removes administrator information.

- **CONCLUSION**

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- The functional dependency analysis of the **SAVANA Online E-Commerce Database Management System** shows that each entity has a clear relationship between its primary key and non-key attributes. Proper normalization reduces redundancy, prevents insertion, update, and deletion anomalies, and improves data integrity, consistency, and efficiency. This database design provides a reliable and well-structured system for managing customers, products, categories, carts, orders, payments, deliveries, and administrators.